Practical 10: Interactive Dashboard, Row-Level Security, and Power Platform Integration

Related topic: Interactive Dashboards, Row-Level Security, Power Apps, and Power Automate (Unit 4)

Name:		Roll No.:		Date:	

Note on figures and data: the tables and visuals in this report were generated from a sample RetailPulse dataset (150 sales orders, 8 products, 8 stores, 4 regions, Jan 2024 - Dec 2025) built to match the lab manual's schema. They reproduce what Power BI Desktop shows for the steps. If your workbook differs, your numbers will differ; paste your own Power BI screenshots into the marked boxes before submitting. Steps 4-6 need the Power BI service and a licensed work/school account; if licences are unavailable the instructor may demonstrate Steps 5 and 6.

Objective

Build an interactive RetailPulse report with drill-through and bookmarks, restrict data by region with Row-Level Security (RLS), and connect the report to Power Automate and Power Apps.

Materials Required

Power BI Desktop, the file from Practical 9, a Power BI service workspace with Pro or Premium Per User licence, Power Automate and Power Apps access through a work or school account, and two test accounts.

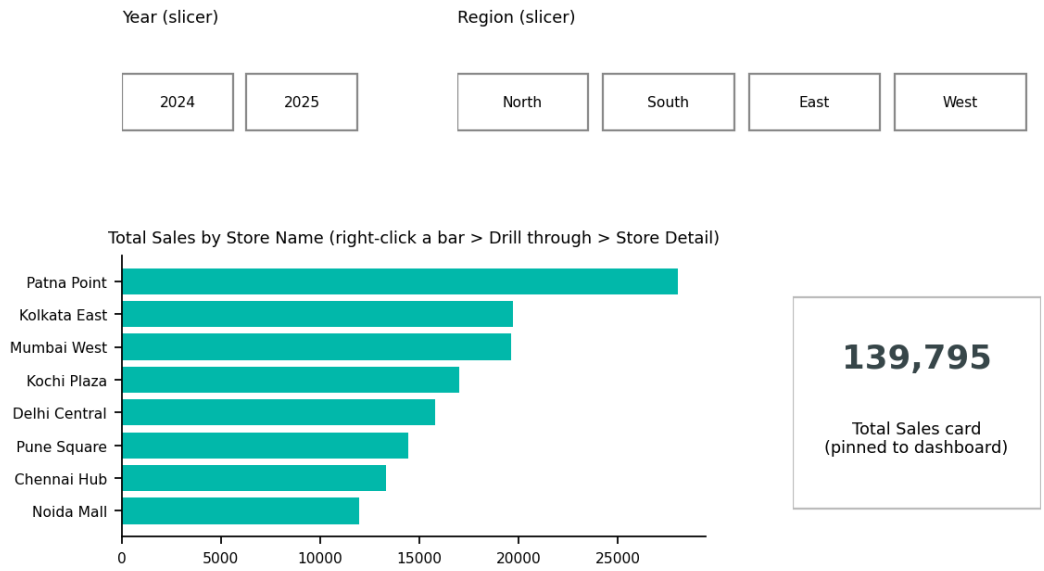
Procedure and Observations

Step 1 - Interactivity.

• **Slicers:** add a slicer with Date[Year] and one with Store[Region]; Format > Slicer settings > Style (tile/dropdown).

• **Drill-through page:** add a new page named **Store Detail**; drag Store[Store Name] to the *Drill through* well of the Visualizations pane; add a card (Total Sales), a table (OrderID, Product Name, Quantity, Sales Amount) and a bar chart. In the main page right-click a store > Drill through > Store Detail. A back button is added automatically.

• **Bookmarks:** on the main page build a chart view and a table view of the same data (stack them and use the Selection pane to show/hide). View > Bookmarks > Add: *Chart View* (table hidden) and *Table View* (chart hidden). Insert > Buttons > Blank, set Action > Type = Bookmark for each button.



Step 2 - Define RLS roles. Modeling > Manage roles > Create. Role **North Manager**, table **Store**, switch to the DAX editor and enter the filter; repeat for **South Manager**:

```
North Manager:  [Region] = "North"      (on table Store)
```

```
South Manager:  [Region] = "South"      (on table Store)
```

Because Store is related to Sales by a one-to-many relationship, filtering Store also filters the Sales rows (Practical 8 model).

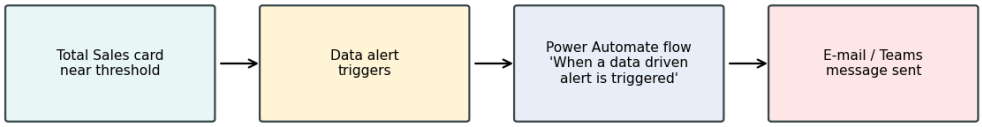
Step 3 - Test with View as. Modeling > View as > tick a role > OK. A yellow bar shows the active role. Results with the sample data:

View as role	Regions visible	Total Sales	Orders	Stores
No role (admin / Desktop)	East, North, South, West	139,794.72	150	8
North Manager	North	27,753.84	34	2
South Manager	South	30,294.89	36	2

The role totals add up to the portion of the full total belonging to North and South; East and West rows are not visible to either manager. Select *Stop viewing* to return to the full view.

Step 4 - Publish and assign users. Home > Publish to the workspace. In the Power BI service: workspace > dataset (semantic model) > ... > **Security** > select *North Manager* > add test account 1 > Add; repeat for *South Manager* with test account 2 > Save. Open the report, hover over the Total Sales card > **Pin** > New dashboard (e.g. RetailPulse Dashboard). Members of the workspace with Admin/Member/Contributor roles bypass RLS, so test with accounts that only have Viewer access.

Step 5 - Alert and Power Automate. On the dashboard, tile ... > **Manage alerts** > Add alert rule: Alert when Total Sales is above a threshold value (e.g. 80% of the current value, so it can be triggered), Notification: *Send me email*. In Power Automate (make.powerautomate.com) create **Automated cloud flow** with trigger **When a data driven alert is triggered (Power BI)**, choose the alert, add the action **Send an email (V2)** or *Post message in a chat or channel (Teams)*, with Subject: *RetailPulse alert* and Body containing the alert title and the tile value from the dynamic content; Save and Test.



Step 6 - Power Apps visual. On the Store Detail page add the **Power Apps** visual (Visualizations pane). Drag Store[Store Name] and Sales[Sales Amount] into *Power Apps data*; choose **Create new** > Power Apps opens with a generated gallery > edit labels if needed > Save and publish the app > return to Power BI. Select a store (via drill-through) and the app shows that store's details. Sample of what appears for the top store:

Store Name	Region	Total Sales	Orders
Patna Point	East	27,998.38	22

OrderID	Product Name	Quantity	Sales Amount
ORD1070	Laptop	5	4,505.70
ORD1077	Phone	5	3,248.24
ORD1138	Phone	4	3,022.34
ORD1132	Laptop	3	2,675.33
ORD1017	Phone	4	2,631.95

Paste your Power BI screenshot here: Drill-through page and bookmark buttons

Paste your Power BI screenshot here: View as role results and Power Automate run history

Result / Expected Outcome Check

• Slicers, drill-through and bookmark buttons work on the report.

• North Manager sees only North-region rows; South Manager sees only South-region rows; the service results match View as.

• The alert triggers the flow and a notification arrives.

• The Power Apps visual updates when a different store is selected.

Viva-Voce Questions and Answers

a) What is Row-Level Security, and how are roles defined in Power BI?

RLS restricts which rows of data a user can see in a report. In Power BI Desktop, roles are defined under Modeling > Manage roles by choosing a table and writing a DAX filter expression, e.g. `[Region] = "North"`. After publishing, users or security groups are assigned to the role in the dataset's Security settings in the Power BI service. Users in a role then see only the rows where the filter is true; filters propagate through relationships to related tables.

b) How can you test RLS?

Use **View as** in Desktop to choose one or more roles (and optionally another user name for dynamic RLS), and use *Test as role* in the service Security page. Confirm the totals change to only that region.

c) What is the difference between static and dynamic RLS?

Static RLS hard-codes the filter value in each role (as here). Dynamic RLS uses `USERPRINCIPALNAME()` against a mapping table, so one role serves all users.

d) What does a drill-through page do?

It gives a detailed page for one selected item (here a store); the selected value passes as a filter to the target page, so all its visuals show only that store.

e) How do Power Automate and Power Apps extend a Power BI report?

Data alerts can trigger Power Automate flows to send e-mails, Teams messages or start workflows; the Power Apps visual embeds an app that receives the selected report context and can show or write back data. Both need licences and a work or school account.